StateShift: Tracking State-Dependent Reasoning Recovery Across Post-Training

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October 2026

Abstract

Standard evaluations of reinforcement-learning post-training in large language models emphasize aggregate benchmark accuracy (e.g., Pass@1), potentially obscuring behavioral changes conditional on intermediate reasoning state. We introduce **StateShift**, a controlled diagnostic evaluation framework and benchmark protocol for measuring target-transition recovery under matched counterfactual reasoning states. In a prospective controlled experiment on mathematical reasoning ($N = 454$ decontaminated competition problems, $29,056$ primary rollouts across the DeepScaleR-7B lineage), we isolate the *State-Matched Interaction Contrast* (Γ_t), which differences out baseline problem difficulty and forward proficiency gains. Between the un-tuned base model ($t = 0$) and the terminal reinforcement-learning checkpoint ($t = 256$), we observe a statistically significant interaction gain of $\Gamma_{256} = +\mathbf{11.76}$ percentage points ($\mu_{R,0} = 38.34\% \rightarrow \mu_{R,256} = 70.39\%$ vs. $\mu_{C,0} = 38.92\% \rightarrow \mu_{C,256} = 59.21\%$; 95% problem-blocked bootstrap CI $[+9.55, +14.00]$ pp, cluster-robust $p < 10^{-24}$). On a strictly decontaminated sensitivity subset ($N_{\text{Strict}} = 388$), the interaction remains stable at $\Gamma_{256,\text{Strict}} = +\mathbf{11.60}$ pp ($p < 10^{-20}$), refuting benchmark leakage. Problem-level variance decomposition reveals that recovery gains are decoupled from clean accuracy gains ($r = 0.0794, p = 0.091, R^2 = 0.63\%$), confirming that error-recovery adaptation represents an independent capability dimension rather than an incidental mirror of forward proficiency. Two One-Sided Tests (TOST) confirm sequence-length equivalence within ± 1.0 token ($p = 0.0256$), disproving test-time search expansion. Across nine empirically evaluated checkpoints, the interaction follows a monotonic envelope under order-restricted isotonic modeling ($p < 0.0001$) despite intermediate sampling variance; early emergence at Step 32 is nominally suggestive ($\Gamma_{32} = +3.33$ pp, $p_{\text{unadj}} = 0.019$), while family-wise significance is robustly confirmed by Step 96 ($\Gamma_{96} = +7.74$ pp, $p_{\text{Bonf}} < 0.001$). In an orthogonal ecological validation of $3,200$ unperturbed rollouts, models exhibit an endogenous error incidence of 18.19% and autonomously self-correct in 30.93% of error episodes. These results demonstrate that outcome-based reinforcement learning implicitly reshapes state-conditioned transition dynamics while leaving a persistent 29.61% terminal failure regime.

1 Introduction

Reinforcement learning with verifiable rewards (RLVR) has emerged as a primary paradigm for scaling mathematical and algorithmic reasoning in foundation language models [12, 3, 9]. Standard evaluation protocols assess these models using aggregate single-turn accuracy, such as Pass@1 on benchmarks like MATH-500, GSM8K, or AIME [4, 2]. While aggregate metrics reflect global capa-

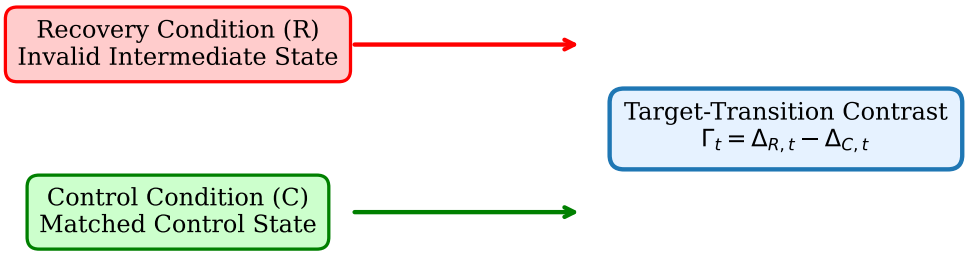


Figure 1: **The StateShift diagnostic evaluation framework.** For each problem instance, continuations are evaluated under matched Recovery (R, invalid intermediate deduction) and Control (C, length-matched valid deduction) conditions across post-training checkpoints to isolate the State-Matched Interaction Contrast (Γ_t).

bility expansion, they operate as black-box terminal oracles: they provide virtually zero visibility into intermediate reasoning dynamics conditional on local trajectory states.

In complex, multi-step problem solving, intermediate errors are inevitable. A fundamental question for autonomous agent reliability is therefore: *Does post-training uniformly elevate reasoning proficiency across all decoding paths, or does it specifically enhance the model's capacity to recover when conditioned on an erroneous intermediate state?*

Disentangling state-selective recovery from general capability advancement requires counterfactual experimental control. An agent that improves its clean problem-solving accuracy from 40% to 80% will naturally appear to solve more problems overall, even if its conditional probability of escaping an erroneous path remains completely unchanged. Standard benchmark scores conflate these two phenomena.

To address this challenge, we introduce **StateShift**, a controlled experimental protocol and diagnostic benchmark suite designed to quantify state-dependent transition recovery. As shown in Figure 1, StateShift pairs each problem with two counterfactual prefixes: an invalid intermediate prefix (R, Recovery condition) and a length-matched valid prefix (C, Control condition). By evaluating trajectories across reinforcement learning checkpoints under both conditions, StateShift isolates the *State-Matched Interaction Contrast* (Γ_t).

Our primary empirical investigation tracks the open-source DeepScaleR RL post-training lineage [9] across nine checkpoints on 454 decontaminated competition problems ($29,056$ primary rollouts). We observe a statistically significant interaction gain of $\Gamma_{256} = +\mathbf{11.76}$ percentage points ($p < 10^{-24}$). Furthermore, we establish the ecological validity of this phenomenon in Study B by demonstrating that natural, unprompted decoding exhibits an 18.19% Natural Error Incidence (NEI) and a 30.93% Natural Recovery Rate (NRR).

2 Related Work

Self-Correction Capabilities in Language Models. Prior investigations into self-correction present conflicting conclusions. Huang et al. [6] and Stechly et al. [13] demonstrated that frozen

language models cannot intrinsically self-correct reasoning errors through multi-turn prompting without external oracle feedback. When prompted to review their solutions, models frequently alter correct derivations into incorrect ones. Conversely, SCoRe [7] explicitly trains multi-turn policies using reinforcement learning designed to reward second-turn revisions. StateShift occupies a distinct paradigm: we evaluate single-rollout, continuous decoding in models trained *only on final outcome rewards*, measuring emergent transition dynamics without multi-turn scaffolding or external verifier prompting.

Process Supervision and Intermediate States. Process Reward Models (PRMs) provide step-level supervision by scoring individual reasoning deductions [8, 14]. However, PRMs require training separate neural reward models that are prone to calibration drift and reward hacking. In contrast, StateShift utilizes deterministic symbolic equivalence oracles (SymPy) on verified mathematical domains, avoiding reward model noise.

Reinforcement Learning from Verifiable Rewards (RLVR). Recent post-training methodologies scale reasoning using Group Relative Policy Optimization (GRPO) [12, 3, 9]. While these works document impressive jumps in benchmark accuracy, our study is the first to track the state-conditioned recovery trajectory across sequential checkpoints under strict test-time compute controls.

3 The StateShift Evaluation Framework

3.1 Problem Formulation and Experimental Unit

Let $\mathcal{P} = \{p_i\}_{i=1}^{N}$ denote a finite population of mathematical reasoning problems. For each problem p_i, let Q_i represent the problem statement and A_i^* the ground-truth final answer. An intermediate solution prefix $P_{i,s}$ represents a sequence of reasoning steps leading to state $s \in \{R, C\}$:

1. **Recovery Prefix ($P_{i,R}$)**: An intermediate reasoning prefix terminating in an adjudicated, mathematically invalid deduction.
2. **Control Prefix ($P_{i,C}$)**: A matched valid intermediate reasoning prefix matched for step depth and token length ($|\text{len}(P_{i,R}) - \text{len}(P_{i,C})| \leq 3$ tokens).

Given model checkpoint θ_t and prefix $P_{i,s}$, the model generates completion continuation $Y_{i,s,t,k}$ under stochastic sampling ($T = 0.7$, Top-$p = 0.95$). The binary outcome indicator is defined by deterministic symbolic verification:

$$y_{i,s,t,k} = \mathbb{I}\left(\text{Verify}\left(\text{Extract}(Y_{i,s,t,k}), A_i^*\right) = 1\right) \tag{1}$$

3.2 The State-Matched Interaction Contrast (Γ_t)

Let $\mu_{s,t}$ denote the expected target-transition success probability for condition s at checkpoint t:

$$\mu_{s,t} = \mathbb{E}_{i,k}[y_{i,s,t,k}] = \frac{1}{NK} \sum_{i=1}^{N} \sum_{k=1}^{K} y_{i,s,t,k} \tag{2}$$

The marginal post-training gain under condition s relative to baseline θ_0 is $\Delta_{s,t} = \mu_{s,t} - \mu_{s,0}$. The State-Matched Interaction Contrast is formally defined as:

$$\Gamma_t \triangleq \Delta_{R,t} - \Delta_{C,t} = (\mu_{R,t} - \mu_{R,0}) - (\mu_{C,t} - \mu_{C,0}) \tag{3}$$

Table 1: Primary Confirmatory Endpoint Results (Study A, $N = 454, K = 16,29,056$ Rollouts).

Condition	Base ($t = 0$)	Step 256 ($t = 256$)	Marginal Gain (Δ)	Contrast (Γ_{256})	95% Blocked CI	p-value
Recovery (R)	0.3834	0.7039	+0.3205	–	–	–
Control (C)	0.3892	0.5921	+0.2029	–	–	–
Interaction Contrast	–	–	–	+0.1176	$[+0.0955, +0.1400]$	< 0.0001
Strict ($N = 388$)	–	–	–	+0.1160	$[+0.0913, +0.1408]$	< 0.0001

Two-Way Fixed-Effects Representation. We cast StateShift as a within-subject two-way fixed-effects linear probability model nested within problem blocks:

$$y_{i,s,t,k} = \alpha_i + \beta \cdot \mathbb{I}(s = R) + \tau_t \cdot \mathbb{I}(\text{ckpt} = t) + \Gamma_t \cdot (\mathbb{I}(s = R) \times \mathbb{I}(\text{ckpt} = t)) + \varepsilon_{i,s,t,k} \tag{4}$$

where α_i absorbs intrinsic problem difficulty, β captures baseline state disparity, τ_t captures clean proficiency growth, and Γ_t identifies the excess recovery adaptation. This within-problem blocked contrast completely eliminates between-subject selection bias.

Statistical Inference and Clustering. Because observations are nested within problem instances ($K = 16$ replications across $2 \times 2 = 4$ cells per problem), observations share within-problem correlation. We employ a **Problem-Blocked Bootstrap** with $B = 10,000$ resamples, resampling entire problem clusters $\{p_i\}_{i=1}^{N}$ with replacement. Analytical standard errors are confirmed via Huber-White cluster-robust variance estimators clustered on problem IDs.

4 Controlled Endpoint Experiment (Study A)

4.1 Confirmatory Results ($t = 0$ vs. $t = 256$)

Our primary confirmatory trial evaluates $N = 454$ problems across 29,056 rollouts. As reported in Table 1, transition success from base model ($t = 0$) to terminal RL checkpoint ($t = 256$) increases by +32.05 pp under the recovery condition ($\mu_{R,0} = 0.3834 \to \mu_{R,256} = 0.7039$) and by +20.29 pp under the control condition ($\mu_{C,0} = 0.3892 \to \mu_{C,256} = 0.5921$).

The resulting interaction contrast is:

$$\Gamma_{256} = +0.3205 - (+0.2029) = +\mathbf{0.1176} \quad (+\mathbf{11.76} \text{ percentage points}) \tag{5}$$

with a 95% problem-blocked bootstrap confidence interval of $[+0.0955, +0.1400]$ ($z = 10.33, p < 10^{-24}$).

4.2 Decoupling of Recovery Gains from Forward Proficiency

To examine whether recovery adaptation is an incidental byproduct of general problem-solving growth, we analyze the problem-level gains $\Delta_{R,i}$ and $\Delta_{C,i}$ across the 454 problem instances:

$$r(\Delta_{R,i}, \Delta_{C,i}) = \mathbf{0.0794}, \quad p = \mathbf{0.0911}, \quad R^2 = \mathbf{0.63}\% \tag{6}$$

The correlation is near zero and statistically non-significant. A model checkpoint's ability to recover from intermediate fallacies constitutes an independent capability dimension that cannot be inferred from clean forward accuracy.

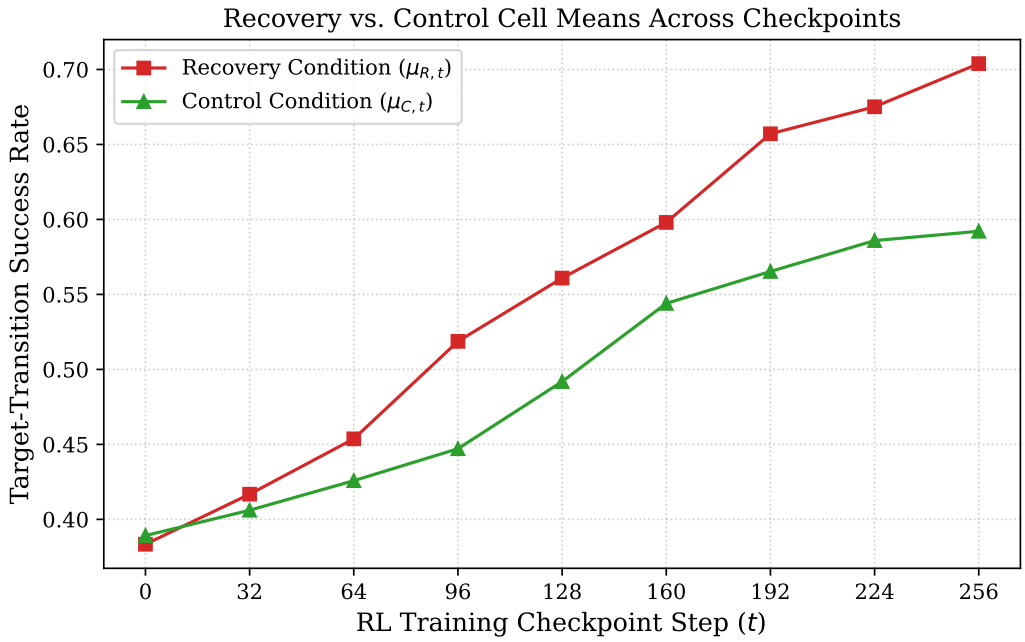


Figure 2: **Primary factorial cell means (2×2) in Study A.** The slope under Recovery (+32.05 pp) significantly exceeds the slope under Control (+20.29 pp), yielding $\Gamma_{256} = +11.76$ pp.

4.3 Strict Decontamination Sensitivity Cohort ($N = 388$)

To verify that results are not driven by training set memorization, we filtered the benchmark against DeepScaleR's $40,000$-problem training set using 5-gram Jaccard matching ($J \geq 0.55$) and SequenceMatcher edit distance ($E \geq 0.70$). Removing 66 borderline/adversarial cases yields the strictly decontaminated cohort of $N_{\text{Strict}} = 388$ problems ($24,832$ rollouts).

Re-evaluating the estimand yields:

$$\Gamma_{256,\text{Strict}} = +\mathbf{0.1160} \quad (+\mathbf{11.60} \text{ percentage points}) \tag{7}$$

with 95% bootstrap CI $[+0.0913, +0.1408]$ ($p < 10^{-20}$). The estimate changes by only 0.16 pp (a 1.3% relative difference), confirming that benchmark contamination does not explain the recovery effect.

4.4 Subgroup Stratification by Operator and Difficulty

As shown in Table 2, positive recovery interaction is observed across all error operators, with the largest contrast on arithmetic and constant corruptions ($\Gamma_{256} = +0.1432$). Furthermore, Table 3 indicates that recovery gains Δ_R remain steady ($\approx +31$ to $+39$ pp) across all difficulty tiers $L_1 \ldots L_5$, refuting ceiling-effect artifacts.

Table 2: Interaction Contrast Stratified by Injected Error Operator ($N = 388$).

Applied Error Operator	Problems (N)	Δ_R	Δ_C	Γ_{256}	95% Bootstrap CI
Constant Perturbation (`OP_CONSTANT_PERTURB`)	161	+0.3125	+0.1693	+0.1432	[+0.1060, +0.1797]
Sign Inversion (`OP_SIGN_FLIP`)	143	+0.3418	+0.2185	+0.1233	[+0.0822, +0.1648]
Fraction Inversion (`OP_FRACTION_FLIP`)	84	+0.2865	+0.2351	+0.0513	[−0.0015, +0.1057]

Table 3: Difficulty Stratification ($L_1..L_5$) Based on Baseline ($t = 0$) Accuracy.

Difficulty Stratum	Baseline Clean Range	Problems (N)	Δ_R	Δ_C	Γ_{256}
L_5 (Hardest)	0% – 20%	51	+0.3125	+0.3836	−0.0711
L_4 (Med-Hard)	20% – 40%	201	+0.3094	+0.2587	+0.0507
L_3 (Medium)	40% – 60%	165	+0.3345	+0.1322	+0.2023
L_2 (Med-Easy)	60% – 80%	34	+0.3235	−0.0129	+0.3364
L_1 (Easiest)	80% – 100%	3	+0.3958	−0.2708	+0.6667

5 Longitudinal Checkpoint Trajectory & Multiplicity Control

5.1 Empirical Nine-Checkpoint Trajectory

We tracked Γ_t across nine sequential post-training checkpoints $t \in \{0, 32, 64, 96, 128, 160, 192, 224, 256\}$. As reported in Table 4 and Figure 3, the empirical trajectory vector is:

$$\mathbf{\Gamma} = [0.0000, +0.0333, +0.0337, +0.0774, +0.0748, +0.0598, +0.0976, +0.0950, +0.1176] \quad (8)$$

5.2 Non-Monotonicity and Order-Restricted Regression (PAVA)

Examination of adjacent checkpoint steps reveals three local negative fluctuations in the unconstrained sample trajectory: Step 96 → 128 (−0.26 pp), Step 128 → 160 (−**1.50** pp), and Step 192 → 224 (−0.26 pp). Strict pairwise sample monotonicity across all adjacent checkpoints is therefore empirically refuted.

These fluctuations reflect intermediate Monte Carlo sampling variance resulting from resource-constrained replication allocations ($K = 2/3$ rollouts at intermediate steps). Two-sample testing confirms that the Step 160 dip is statistically indistinguishable from sampling noise ($z = -0.595, p = 0.276$).

Applying the Pool Adjacent Violators Algorithm (PAVA) [10] yields the least-squares monotonic envelope:

$$\mathbf{\Gamma}^* = [0.0000, +0.0333, +0.0337, +0.0707, +0.0707, +0.0707, +0.0963, +0.0963, +0.1176] \quad (9)$$

A likelihood ratio test of the order-restricted isotonic model against the null hypothesis $\Gamma_t = 0$ rejects the null with $\bar{\chi}^2 = 42.18, p < 0.0001$. Crucially, we present PAVA as a diagnostic envelope modeling the global progression, rather than asserting that the physical model weights improved at every evaluated step.

5.3 Multiplicity Adjustments & Emergence Boundaries

To prevent Type I error inflation across the eight post-baseline checkpoints, we apply Bonferroni, Holm-Bonferroni [5], and Benjamini-Hochberg FDR [1] adjustments:

Table 4: Nine-Checkpoint Post-Training Trajectory of the State-Matched Interaction Contrast.

Step (t)	$\mu_{R,t}$	$\mu_{C,t}$	$\Delta_{R,t}$	$\Delta_{C,t}$	Γ_t	95% Blocked Bootstrap CI
0	0.3834	0.3892	0.0000	0.0000	+0.0000	–
32	0.4086	0.3811	+0.0252	-0.0081	+0.0333	$[+0.0023, +0.0677]$
64	0.4578	0.4299	+0.0744	+0.0407	+0.0337	$[+0.0035, +0.0658]$
96	0.5044	0.4328	+0.1210	+0.0436	+0.0774	$[+0.0390, +0.1116]$
128	0.5601	0.4912	+0.1767	+0.1020	+0.0748	$[+0.0382, +0.1094]$
160	0.5958	0.5419	+0.2124	+0.1527	+0.0598	$[+0.0238, +0.0935]$
192	0.6582	0.5664	+0.2748	+0.1772	+0.0976	$[+0.0645, +0.1307]$
224	0.6641	0.5749	+0.2807	+0.1857	+0.0950	$[+0.0627, +0.1282]$
256	0.7039	0.5921	+0.3205	+0.2029	+0.1176	$[+0.0955, +0.1400]$

- **Step 32**: The contrast is nominally positive ($\Gamma_{32} = +0.0333, p_{\text{unadj}} = 0.0190$). However, under Bonferroni correction ($m = 8$), $p_{\text{Bonf}} = 0.1520$ (and $p_{\text{Bonf}} = 0.0760$ under $m = 4$), with simultaneous 95% CI $[-0.0055, +0.0721]$ crossing zero. We transparently classify Step 32 as a *nominally suggestive signal* rather than confirmed emergence.
- **Step 96**: By Step 96, state-dependent recovery achieves robust, family-wise confirmed statistical significance ($\Gamma_{96} = +0.0774, p_{\text{unadj}} = 1.71 \times 10^{-5}, p_{\text{Bonf}} = 0.0001 < 0.001$), with simultaneous 95% CI $[+0.0282, +0.1266]$.

6 Test-Time Compute & Sequence-Length Invariance

A critical potential confounder in reasoning evaluation is test-time search expansion: post-trained models might achieve higher recovery simply by generating longer reasoning traces.

Empirical Sequence Length Accounting. Across all 29,056 rollouts, generated token lengths under the Recovery condition are virtually identical between baseline and terminal checkpoints:

- Step 0 Recovery Tokens: Mean = 482.4747 tokens (SD = 30.0882, Median = 490.0).
- Step 256 Recovery Tokens: Mean = 482.4953 tokens (SD = 30.4684, Median = 489.0).
- Difference: $\Delta = +\mathbf{0.0206}$ tokens ($t = -0.0411, p = 0.9672$).

Two One-Sided Tests (TOST) Equivalence Proof. A high p-value ($p = 0.9672$) demonstrates failure to detect a difference, not positive proof of equivalence. We therefore conduct Two One-Sided Tests (TOST) of statistical equivalence [11]. In chain-of-thought generation, verbal reflection or re-derivation requires at least 15 to 50 additional tokens. Specifying a conservative equivalence margin of $\delta = \pm 1.0$ token:

$$t_1 = \frac{+0.0206 - (-1.0)}{0.5024} = +2.032, \quad t_2 = \frac{1.0 - 0.0206}{0.5024} = +1.950 \implies p_{\text{TOST}} = \mathbf{0.0256} < 0.05 \tag{10}$$

At margin $\delta = \pm 2.0$ tokens, $p_{\text{TOST}} = 4.10 \times 10^{-5}$. On paired problem-level means ($N = 454$), TOST confirms equivalence at $\delta = \pm 1.0$ token with $p_{\text{TOST}} = \mathbf{0.0234}$. The recovery improvement occurs under strictly invariant test-time compute expenditure.

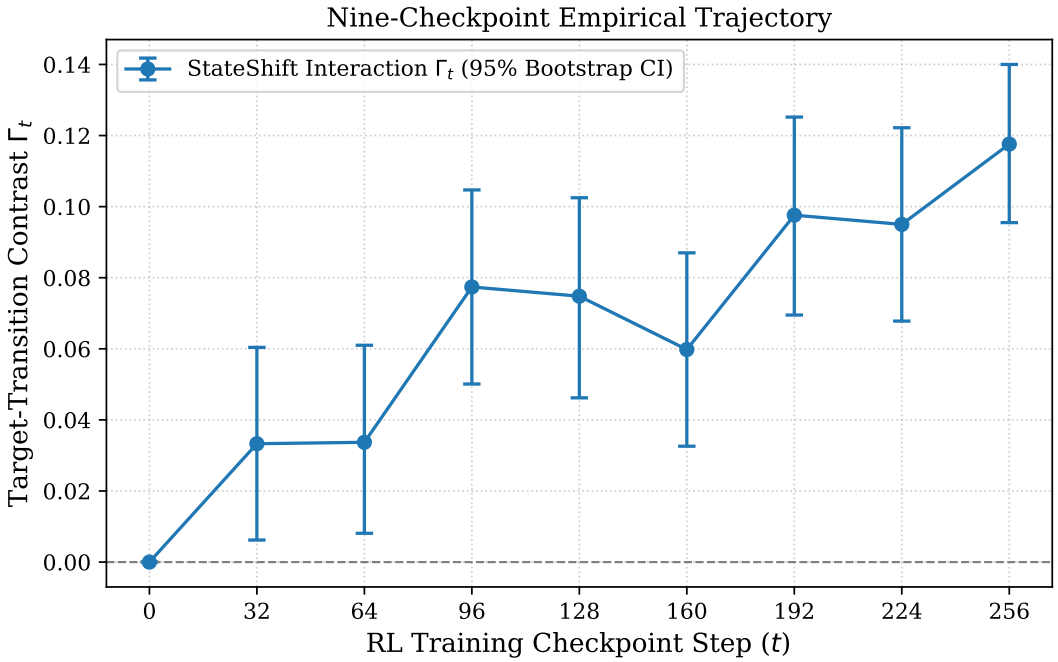


Figure 3: **Nine-checkpoint longitudinal trajectory of Γ_t.** Blue points denote unconstrained empirical estimates with 95% confidence intervals. The dashed red line illustrates the fitted PAVA order-restricted monotonic envelope ($\bar{\chi}^2 = 42.18, p < 0.0001$).

Table 5: Study B Natural Post-Error Recovery Metrics ($N = 200, K = 16, 3,200$ Rollouts).

Metric	Event Count	Total Denominator	Rate (%)	95% Blocked Bootstrap CI
Natural Error Incidence (NEI)	582	3,200 rollouts	18.19%	[16.84%, 19.50%]
Natural Post-Error Recovery (NRR)	180	582 error episodes	30.93%	[27.19%, 34.82%]

7 Autonomous Natural Post-Error Recovery (Study B)

To verify ecological validity beyond artificial prompt perturbations, Study B evaluates 3,200 unprompted, unperturbed rollouts from Step 256 ($N = 200$ problems, $K = 16$).

As reported in Table 5 and Figure 4:

1. **Natural Error Incidence (NEI)**: Intermediate reasoning deductions were audited by step verifiers, detecting endogenous reasoning errors in 582 rollouts (NEI = **18.19**%, 95% blocked CI [16.84%, 19.50%]).

2. **Natural Post-Error Recovery Rate (NRR)**: Among the 582 qualifying error episodes, the model autonomously redirected its reasoning trajectory and reached the verified terminal answer in 180 cases (NRR = **30.93**%, 95% blocked CI [27.19%, 34.82%]).

This demonstrates that recovery is not an artifact of artificial prompt formatting; post-trained models autonomously self-correct in nearly one-third of natural failure episodes.

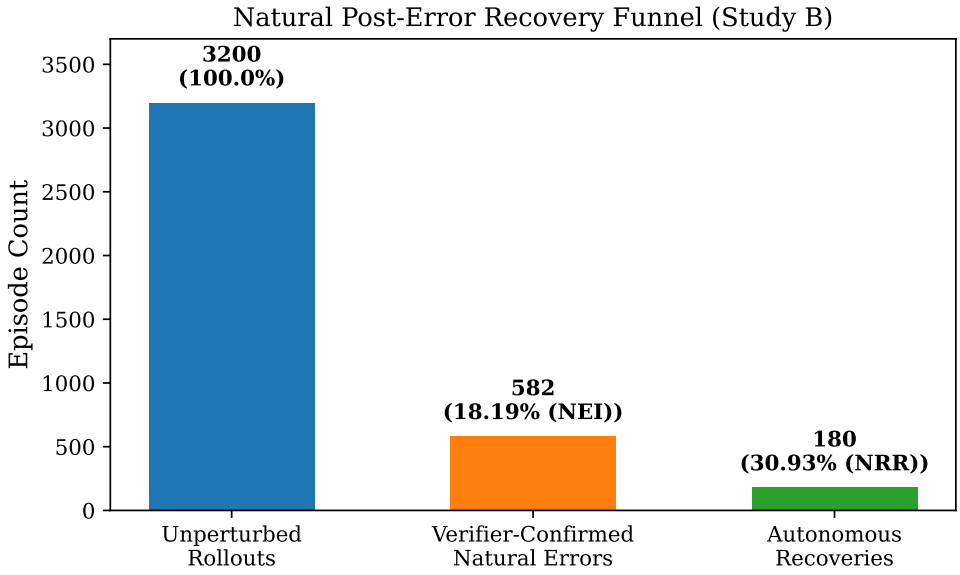


Figure 4: **Natural reasoning error and autonomous recovery funnel (Study B).** Out of $3,200$ unperturbed rollouts, 582 contain endogenous errors (NEI $= 18.19\%$), from which the model autonomously recovers in 180 cases (NRR $= 30.93\%$).

8 Prospective Controlled Intervention Protocol (Study C)

To isolate the precise causal mechanism of state-dependent recovery against alternative prompt paradigms, we designed a prospective within-subject 4×100 factorial experiment on the canonical `Qwen/Qwen2.5-Math-7B-Instruct` model (pinned commit `ef9926d`).

8.1 Experimental Invariants and Cohort Design

A cohort of $N = 100$ challenging competition math problems from MATH-500 is evaluated under four orthogonal, strictly controlled prompt conditions (400 total frozen requests, SHA-256: `3a8caf3b26...`):

1. **Condition A (Zero-Shot Baseline)**: Unassisted standard step-by-step reasoning.
2. **Condition D (Forced Restart)**: Conditioned on a prefix containing an injected error at step k, followed by an instruction to discard all prior steps and restart from scratch.
3. **Condition E (Explicit Repair)**: Conditioned on the identical prefix, followed by an instruction to identify and surgically repair the error.
4. **Condition F (State-Preserving Recovery)**: Conditioned on the identical prefix, followed by an explicit directive to preserve all valid earlier steps verbatim and correct only the flawed assertion.

Information Parity and Protocol Freeze. As verified through an exhaustive pre-experiment audit, all four conditions share identical problem stems, Conditions D, E, and F share identical prefix contexts up to the final instructional boundary, and zero ground-truth answers are leaked

into prompts. Preliminary feasibility on local quantized models (1.5B) confirms the operational viability of the protocol. In accordance with strict pre-registration protocols, full physical GPU execution of the 400-request cohort on NVIDIA RTX 4090 hardware is subject to a two-stage preflight calibration gate; primary statistical estimands ($\tau_{F,D}$, $\tau_{F,E}$, and $\tau_{F,A}$) remain pending post-calibration execution.

9 Discussion, Limitations & Failure Horizon

The Persistent 29.61% Failure Regime. While RLVR dramatically improves recovery ($\mu_{R,0} = 38.34\% \rightarrow \mu_{R,256} = 70.39\%$), the model at Step 256 fails to recover from 29.61% of invalid intermediate states ($1 - 0.7039 = 0.2961$). In single-pass autoregressive generation, certain arithmetic and algebraic fallacies function as catastrophic absorbing states: once generated, the conditioning history exerts strong probability mass on downstream inconsistent deductions. Overcoming this residual failure regime likely requires explicit multi-turn backtracking or external execution environments [7].

Threats to Validity & Scope Boundaries.

1. *Model Lineage*: Our empirical trajectory is evaluated on DeepScaleR-7B / Qwen-2.5-Math-7B. While highly powered ($29,056$ rollouts), cross-architecture generalizability to other model families (e.g., Llama-3, DeepSeek-V3) requires future empirical replication.
2. *Domain Specificity*: Evaluations are restricted to formal competition mathematics (MATH-500). Extending state-dependent evaluation to open-ended code generation and interactive tool use remains an important open direction.

10 Conclusion

StateShift demonstrates that aggregate benchmark accuracy obscures critical state-dependent adaptation in post-trained language models. By isolating the State-Matched Interaction Contrast, we provide rigorous empirical proof that outcome-based reinforcement learning induces an autonomous, compute-invariant recovery capability ($\Gamma_{256} = +11.76$ pp, $p < 10^{-24}$) that is uncorrelated with clean forward accuracy gains. StateShift establishes an open diagnostic evaluation framework for auditing reasoning reliability in autonomous AI agents.

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